

Fairness-Aware Service Allocation in Adaptive Quantum Networks: Risks, Metrics, and Auditable Mediation

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Abstract—This reference analysis examines fairness in a quantum-network controller that selects entanglement routes and allocates scarce qubits under changing threat conditions. The central concern is allocative: an efficiency-oriented bandit policy may deliver reliable service to some flow classes while concentrating delay, retries, or starvation on others. The proposed response is not a claim that one fairness statistic solves the problem. It is a measurement-first workflow that defines service classes before evaluation, records demand and resource access, compares an unmediated policy with an auditable mediator, and reports the resulting utility–disparity frontier with explicit residual risk.

I. SYSTEM CONTEXT AND RESPONSIBILITY BOUNDARY

The artifact under analysis is a threat-aware quantum-network controller, not a generic classifier. At each decision round, the controller observes a network state, selects an entanglement route, and allocates limited qubit-generation capacity. Link quality, memory constraints, replay capacity, allocator choice, and adversarial conditions can change the quality of the available actions. Prior routing work supplies the inherited bandit and threat-regime evaluation spine; the DSCI 602 contribution is a planned fairness-aware mediation layer for quantum service equity. Routing and resource allocation therefore remain the primary technical problem, while clinical applications are future context only.

The system influences who receives successful and timely quantum service under scarcity. Relevant stakeholders are applications or tenants competing for flows, network operators responsible for allocation policy, and downstream users who depend on the resulting entanglement paths. The initial equity units should be operational service classes defined before inspecting outcomes, such as demand tiers, latency requirements, or context-quality cohorts. These are not assumed to be demographic groups; if a future deployment maps them to people or institutions, protected-attribute analysis and stakeholder review would be required.

This boundary matters because a network-wide efficiency score can conceal service differences. Inherited configuration-specific evidence includes a Thompson-sampling replay slice whose Oracle-normalized efficiency changes from 89.2% to 84.9% and then 90.8% as replay capacity increases under an online-adaptive threat. That sequence is not a DSCI 602 fairness result. It is an example of why aggregate behavior deserves disaggregation: a gain or loss may be distributed unevenly

across flow classes. Quantum-network routing and fair-sharing research likewise treats scarce capacity and simultaneous requests as allocation questions rather than merely throughput questions [1], [2].

II. WHERE FAIRNESS RISK ENTERS

Suresh and Gutttag’s life-cycle framework provides a useful map for this pipeline [3]. Representation harm can occur when evaluation omits difficult topologies, low-context flows, or relevant service classes. Measurement harm occurs when traces record success and reward but omit demand, allocated qubits, route choice, fidelity, retries, starvation age, or context quality. Aggregation harm occurs when one reward treats incompatible service needs as interchangeable. Evaluation harm occurs when aggregate efficiency is reported without class-level outcomes. Deployment harm occurs when a policy tuned at one threat regime is transferred to another operating point.

Sequential learning adds a feedback-loop risk. A bandit observes evidence only for selected actions. If a route or class receives fewer opportunities, it can produce less reward evidence, which can reduce its future selection probability. Fairness constraints therefore have a learning and regret cost, rather than being a free post-processing step [4]. Fair contextual bandit work similarly motivates minimum service rates and explicit treatment of context when outcomes differ across users [5]. In this system, poor context can become an allocation disadvantage even when no protected attribute is used.

The plausible harms are concrete. High-demand or high-degree flows may receive most successful allocations while low-demand flows absorb waiting and retries. Flows with weak context may be treated as unreliable and consequently receive less exploration. Long or low-fidelity routes may be repeatedly deprioritized, creating starvation that disappears inside a network-wide mean. These are allocative harms: access to a scarce capability is withheld or degraded. They also create accountability harms if operators cannot reconstruct why a flow received inferior service. The general fairness literature shows why aggregate performance is insufficient: class-level failures can be hidden by overall accuracy or a seemingly neutral proxy [6], [7].

III. FAIRNESS OBJECTIVE AND MEASUREMENT

Classical definitions are useful warnings but cannot be copied mechanically into a resource-allocation system. Demographic parity would compare success or access rates across classes, while equalized odds and equality of opportunity compare error behavior conditional on an outcome [8]. Calibration and parity can conflict under unequal base rates, and impossibility results show that several desirable criteria cannot generally hold simultaneously [9], [10]. The responsible objective is therefore to report a frontier rather than declare one scalar to be “fair.”

One candidate primary measure is demand-normalized access. For class g over a run horizon T , let $Q_{g,t}$ be allocated quantum capacity and $D_{g,t}$ be declared demand:

$$A_g(T) = \frac{\sum_{t=1}^T Q_{g,t}}{\sum_{t=1}^T D_{g,t}}, \quad \Delta_A(T) = \max_{g,h} |A_g(T) - A_h(T)|. \quad (1)$$

The audit should also report success-rate and latency gaps, worst-class outcomes, retry and starvation frequency, fidelity failures, and a bounded resource-allocation index such as Jain’s index [11]. Zero-demand classes require a declared exclusion rule, and every class definition, denominator, physical-infeasibility condition, and missing-context rule should be versioned with the experiment. The point is not to hide complexity inside a single index; it is to make the trade-offs inspectable.

IV. MEDIATION AND EVALUATION PROTOCOL

The first mitigation is measurement. An enriched decision trace should record the class definition, demand, context-quality indicators, candidate routes, policy scores, selected route, allocated qubits, fidelity estimate, success, latency, retries, and starvation age. Constructed unit tests should include equal service, unequal service, zero demand, missing context, and physically infeasible routes. This is a prerequisite for any claim about disparity.

The second mitigation is an in-processing mediator attached at a documented policy-score or allocator boundary. A candidate score can be written as

$$S_t^{\text{fair}}(a) = U_t(a) - \lambda[\hat{\Delta}_t(a) - \epsilon]_+ - \gamma C_t(a), \quad (2)$$

where U_t is base utility, $\hat{\Delta}_t$ is estimated disparity, ϵ is tolerated slack, and C_t penalizes missing or unreliable context. The identity setting $\lambda = \gamma = 0$ is an explicit off-switch, not an afterthought. The mediator should remain separable and auditable so that the inherited routing policy can be compared with its fairness-aware counterpart [5].

Validation must compare the unmediated baseline with a prespecified sweep of λ , γ , and ϵ across matched policies, allocators, capacities, topologies, threat regimes, and random seeds. Report aggregate utility, regret, success, and fidelity beside Δ_A , latency gaps, worst-class outcomes, Jain’s index, and starvation frequency. Repeated trials and uncertainty intervals are necessary because small classes can produce unstable estimates. The result should be a utility–disparity

frontier, not a single favorable operating point selected after inspection. If a class has too few observations for a stable estimate, the system should report uncertainty and apply a conservative service floor rather than infer that no disparity exists.

V. LIMITATIONS AND RESIDUAL RISK

This analysis is prospective. It does not claim completed live mediation or completed DSCI 602 fairness experiments. The simulator does not establish hardware-grounded timing, contention, or noise behavior. Service classes are design choices and may encode the wrong units of concern; demand itself may be strategically inflated. A mediator can reduce utility, increase latency for other classes, or favor an imperfect proxy for equity. Conversely, removing mediation can preserve aggregate utility while hiding persistent exclusion.

The remaining governance questions are material: who approves classes and denominators, who reviews alerts, who can change fairness weights, and what rollback path exists? Security and privacy intersections, including poisoning of threat feedback and exposure of flow telemetry, require separate analysis. The responsible claim is narrow: adaptive quantum routing can create service inequity while appearing efficient, so the project should measure flow-conditioned outcomes, evaluate an auditable mediator, and disclose the frontier and its residual uncertainty.

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